



Tutorial 1

The core exercises are considered part of the course material, and you are advised to at least finish these. For additional practice and interesting applications, advanced exercises are available as well – feel free to pick and choose from these.

Unless otherwise specified, we expect you to create your own algorithms for the exercises in each tutorial. This means that you should not use libraries (e.g. SciPy) for the methods that are part of the course material.

Core exercises

1. FFT and effect of aliasing

In this exercise, you will see how insufficient sampling can significantly alter a signal while doing FFT because of aliasing.

- Write a function that performs the Fast Fourier Transform on a one-dimensional array.
- Consider the function $f(x) = (2x + \sin(2\pi x/5) + 3\cos(2\pi x/2))\sin 2x$. Take 8 equally spaced points in the range $x = [0, 20]$ and evaluate the FFT coefficients for the function values at these points. Plot the amplitudes of the FFT coefficients $F(k)$ vs Fourier space variable k . Use `pyplot.stem()` to better display the frequency components.
- Implement an inverse FFT. Reconstruct $f(x)$ and plot it together with the original signal. How does the reconstruction look?
- Do the same as above, but for 16, 32, 64, 128, and 256 points (or any meaningful subset). How do the Fourier components and reconstructed functions change with the number of points? How does the performance change if you take 95 points instead?
- [*Bonus*] Separate the sinusoidal components of the function from the total value and overplot the components along with the function. Now look at the Fourier domain range for the FFT with the smallest and largest number of points from the previous point. Take three most dominant frequency components from each of those and reconstruct the function for each case. How do they compare with each other and the original function?

2. Tiling and the roles of specific FT modes

Like discussed in the lectures, the Fourier transform assumes its input to be periodic, which for an image means that it is assumed to tile well. When this is not the case, the Fourier transform has significant power added along axes which have discontinuities. In this exercise we are going to take a closer look at this additional power and what it does. We are also going to remove contamination using the FFT.

- Download the wolf image from the lectures:
`https://home.strw.leidenuniv.nl/~daalen/files/wolf.jpg`
Use e.g. `scipy.ndimage.imread` to read it and `pyplot` to plot it. Tile the image (i.e. plot it 2x2) and note where it is not periodic. Take its FFT and plot the log of its magnitude (absolute value) as an image. Remember that the order of the modes along each axis is $[0, 1, \dots, f_c, -f_c + 1, -f_c + 2, \dots, -1]$! It's usually a good idea to shift the output such that mode (0,0) is in the center instead of the bottom left, but don't forget to shift back before taking the inverse FFT. Is there additional power where you expect it?
- Remove the additional power as well as you can in Fourier space, such that when you plot it again the line of additional power now blends in well with the rest of the Fourier transform. Make sure you edit both the real and imaginary part. Now take the inverse FFT and tile the result. Did the wolf image change in such a way so as to tile better?
- [*Bonus*] Now download another image:
`https://home.strw.leidenuniv.nl/~daalen/files/cat_contam.jpg`
This picture of a cat in the dark has several contaminated modes. Identify and clean these in Fourier space, then transform back to see if all contaminants were removed.



- (d) *[Bonus]* The FFT of the cat picture has two striking “fans” at $k_1 < 0, k_2 > 0$ and $k_1 > 0, k_2 < 0$. What features in the original image cause these?

Advanced exercises

3. Rotation and translation of images under FFT

Fourier transforms have a lot of applications in astrophysics and other fields of science. In this question we will look at some simple examples of two dimensional images and do a Fourier transform on these images. Also we will look at how the Fourier transform changes under rotation and translation.

- (a) As a first step write down the definition of the periodic Fourier transform and the periodic discrete Fourier transform in two dimensions.
- (b) Construct an image of a two dimensional Gaussian centered at the center of your image. Assume an image size of 512×512 pixels, and $\sigma = 60$ pixels for the Gaussian. Calculate the Fourier transform of this two dimensional image.
- (c) Make another image of rectangle with a constant height, and size $a \times b$ and calculate the Fourier transform of this image. Assume an image size of 512×512 pixels with value 0 in all pixels, except within the rectangle, where the value is the height that you choose (any value different than 0 would work). What happens if $a = b$ (assume $a = b = 300$ pixels) and what happens if $a > b$ (assume $a = 300$, and $b = 100$ pixels)?
- (d) Make an image (512×512 pixels) with a centered disk in the center with a constant height (again, you can choose any value different than 0) and a radius $r = 200$ pixels. Calculate the Fourier transform of this image.
- (e) Make an image with a two pixels displaced from the center with equal distance but on opposite sides and calculate the Fourier transform. What happens if you change the distance between both points?
- (f) In general, do a rotation on the result of part a, and determine whether Fourier transforms are invariant under Fourier transforms.
- (g) Do the same but now for a general translation.
- (h) Now do a discrete Fourier transform of the image which can be downloaded here: <https://home.strw.leidenuniv.nl/~garcia/NUR/darkside.jpeg>. Make a plot of the phases and amplitudes of this image.
- (i) Calculate the phases and amplitudes in the case that we rotate the previous image under different angles. Plot the phases and amplitudes for different angles.
- (j) Do the same but for different values of the translation both in the x axis and the y axis, use a translation with periodic conditions.
- (k) Construct a way of determining the FFT of an image by making it invariant under translations, and check your code.
- (l) Is it possible to construct an invariant FFT under rotations? If so construct an FFT which is invariant under rotations.